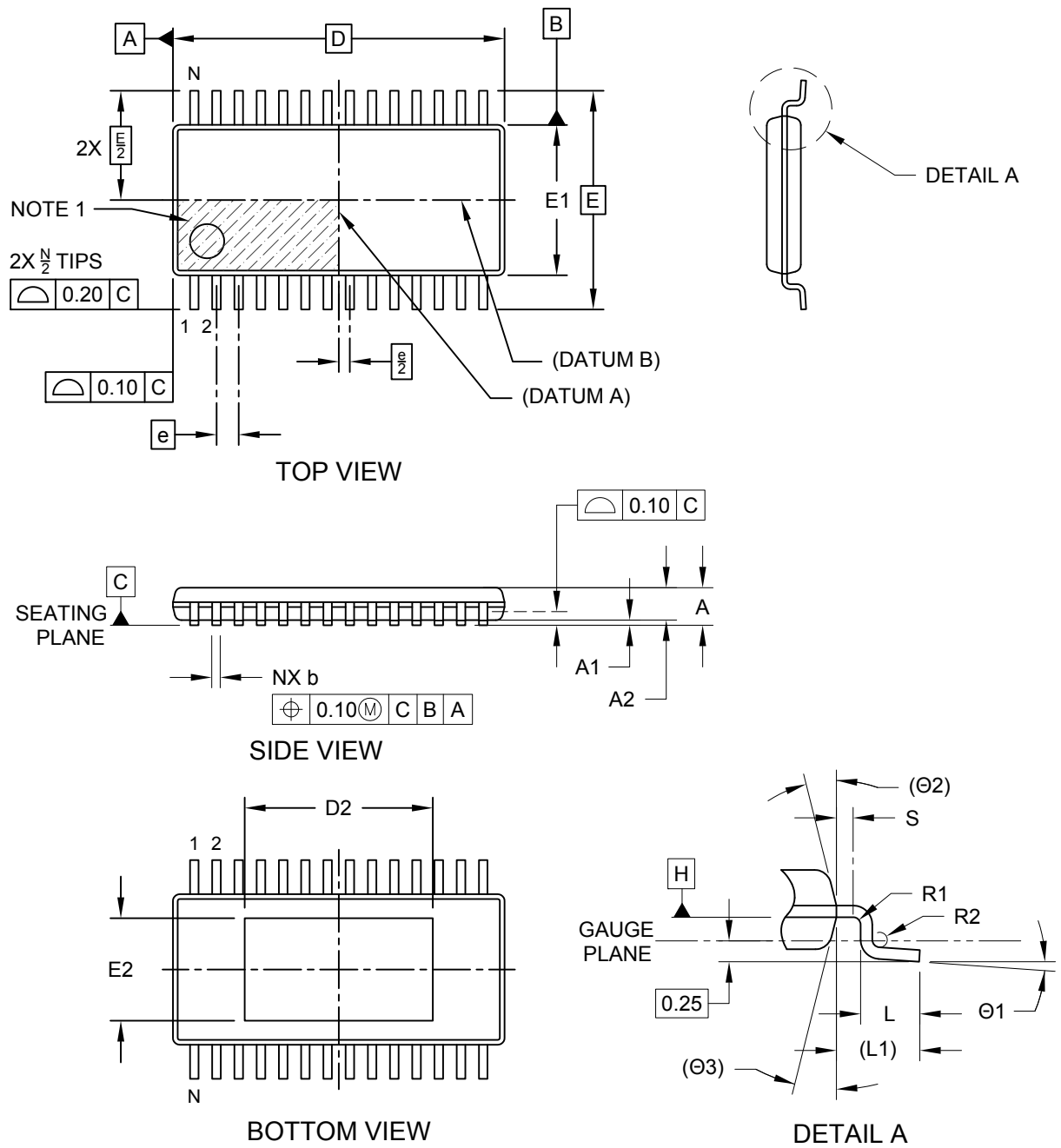


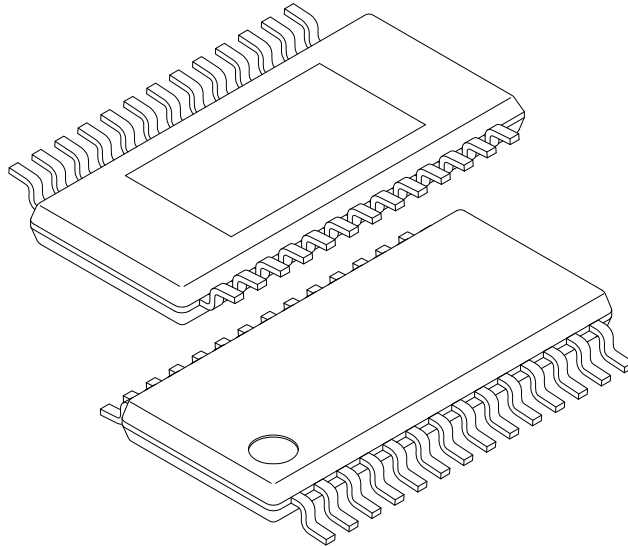
# **28-Lead Thin Shrink Small Outline Package (ST) - 4.4 mm Body [TSSOP]** **With 5.5x3.0 mm Exposed Pad**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 28-Lead Thin Shrink Small Outline Package (ST) - 4.4 mm Body [TSSOP] With 5.5x3.0 mm Exposed Pad

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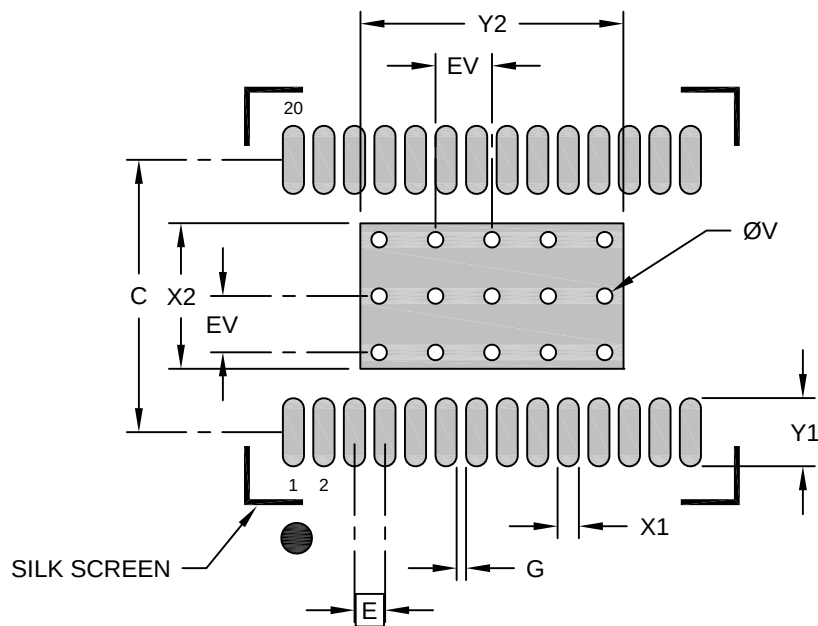
| Units                |    | MILLIMETERS |      |      |
|----------------------|----|-------------|------|------|
| Dimension Limits     |    | MIN         | NOM  | MAX  |
| Number of Terminals  | N  | 28          |      |      |
| Pitch                | e  | 0.65 BSC    |      |      |
| Overall Height       | A  | -           | -    | 1.10 |
| Standoff             | A1 | 0.05        | -    | 0.15 |
| Terminal Thickness   | A2 | 0.85        | 0.90 | 0.95 |
| Overall Length       | D  | 9.70 BSC    |      |      |
| Exposed Pad Length   | D2 | 5.40        | 5.50 | 5.60 |
| Overall Width        | E  | 6.40 BSC    |      |      |
| Molded Package Width | E1 | 4.30        | 4.40 | 4.50 |
| Exposed Pad Width    | E2 | 2.90        | 3.00 | 3.10 |
| Terminal Width       | b  | 0.19        | -    | 0.30 |
| Terminal Length      | L  | 0.50        | 0.60 | 0.70 |
| Terminal Shoulder    | S  | 0.20        | -    | -    |
| Terminal Bend Radius | R1 | 0.09        | -    | -    |
| Terminal Bend Radius | R2 | 0.09        | -    | -    |
| Terminal Foot Angle  | Θ1 | 0°          | -    | 8°   |
| Mold Draft Angle     | Θ2 | 14° REF     |      |      |
| Mold Draft Angle     | Θ3 | 14° REF     |      |      |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

## 28-Lead Thin Shrink Small Outline Package (ST) - 4.4 mm Body [TSSOP] With 5.5x3.0 mm Exposed Pad

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Contact Pitch            | E  | 0.65 BSC    |      |      |
| Center Pad Width         | X2 |             |      | 3.10 |
| Center Pad Length        | Y2 |             |      | 5.60 |
| Contact Pad Spacing      | C  |             | 5.80 |      |
| Contact Pad Width (X28)  | X1 |             |      | 0.45 |
| Contact Pad Length (X28) | Y1 |             |      | 1.45 |
| Contact Pad to Pad (X26) | G1 | 0.20        |      |      |
| Thermal Via Diameter     | V  |             | 0.33 |      |
| Thermal Via Pitch        | EV |             | 1.20 |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process